

Math 10B with Professor Stankova

Quiz 1; Tuesday, 1/23/2018

Section #203; Time: 9:30 AM

GSI name: Roy Zhao

Name: _____

Circle True or False or leave blank. (1 point for correct answer, -1 for incorrect answer, 0 if left blank)

1. **TRUE** False Sometimes, we have to use both the product rule and the sum rule to solve a problem.
2. True **FALSE** Whenever the problem says “at least”, then we have to use complementary counting.

Solution: If the problem says, how many ways can we flip 100 coins and get at least 100 heads, then it is clear that we should do this via complementary counting.

Show your work and justify your answers. Please circle or box your final answer.

3. (10 points) (a) (3 points) How many different license plates are there if a license plate consists of 3 letters followed by 3 numbers or consists of 6 numbers.

Solution: $26^3 \cdot 10^3 + 10^6$.

- (b) (4 points) How many numbers less than or equal to 100 are divisible by 4 or 5?

Solution: There are $100/4 = 25$ numbers divisible by 4, there are $100/5 = 20$ numbers divisible by 5, and $100/20 = 5$ numbers divisible by both. So, there are a total of $25 + 20 - 5 = 40$ numbers divisible by either.

- (c) (3 points) How many ways are there to flip 5 coins and get at least 1 head?

Solution: The complementary case is flipping and getting less than 1 head, which can only occur by getting 0 heads. This can only happen in 1 way. Thus, there are $2^5 - 1 = 31$ different ways to get at least 1 head.